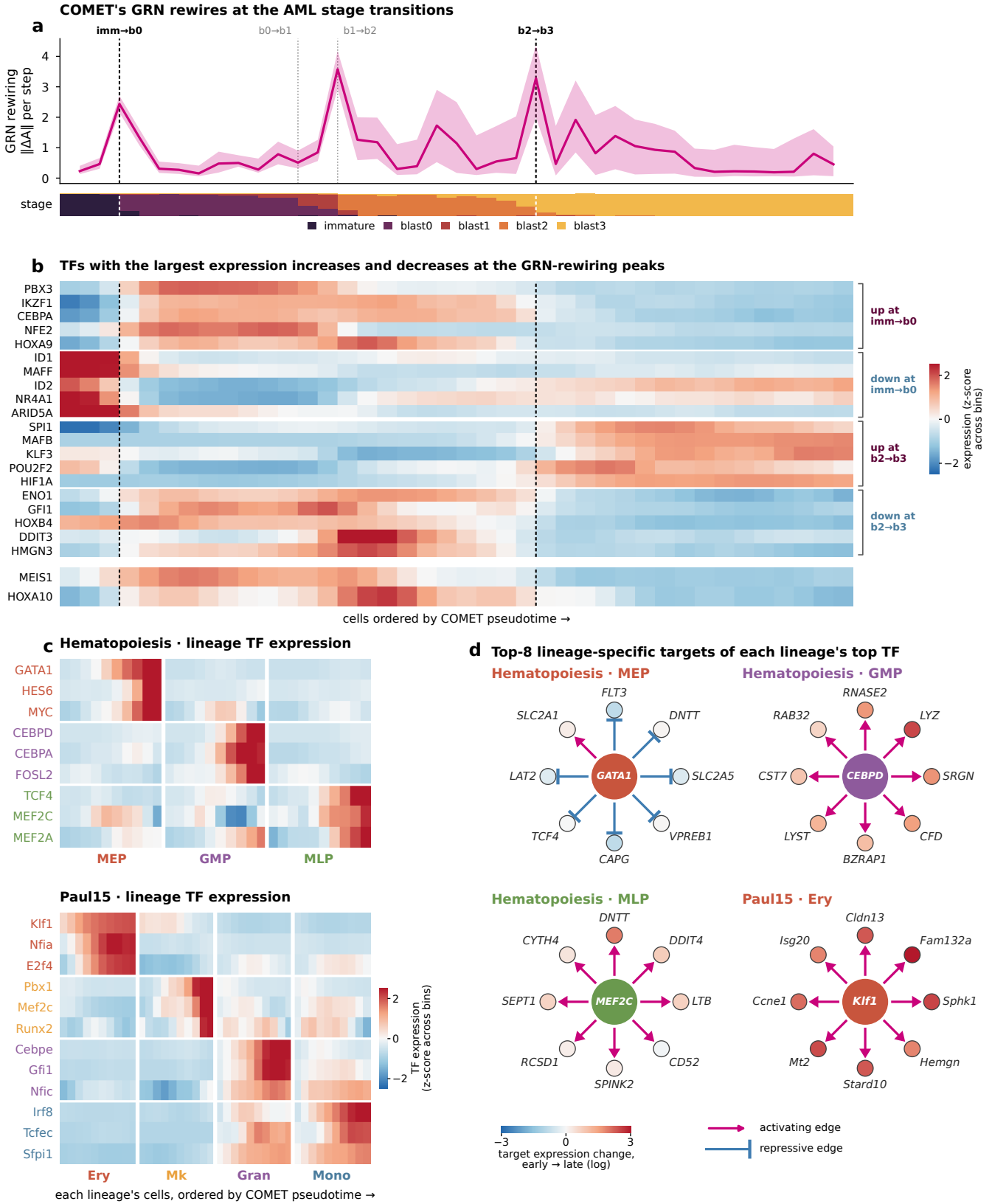

COMET's GRN rewires at the AML stage transitions
a
imm→b0
b0→b1
b1→b2
b2→b3
GRN rewiring
‖ΔA‖ per step
stage
immature
blast0
blast1
blast2
blast3
b
TFs with the largest expression increases and decreases at the GRN-rewiring peaks
PBX3
IKZF1
CEBPA
NFE2
HOXA9
ID1
MAFF
ID2
NR4A1
ARID5A
SPI1
MAFB
KLF3
POU2F2
HIF1A
ENO1
GFI1
HOXB4
DDIT3
HMGN3
MEIS1
HOXA10
up at imm→b0
down at imm→b0
up at b2→b3
down at b2→b3
expression (z-score across bins)
cells ordered by COMET pseudotime →
c
Hematopoiesis · lineage TF expression
GATA1
HES6
MYC
CEBPD
CEBPA
FOSL2
TCF4
MEF2C
MEF2A
MEP
GMP
MLP
Paul15 · lineage TF expression
Klf1
Nfia
E2f4
Pbx1
Mef2c
Runx2
Cebpe
Gfi1
Nfic
Irf8
Tcfec
Sfpi1
Ery
Mk
Gran
Mono
TF expression (z-score across bins)
each lineage's cells, ordered by COMET pseudotime →
d
Top-8 lineage-specific targets of each lineage's top TF
Hematopoiesis · MEP
FLT3
SLC2A1
DNTT
LAT2
GATA1
SLC2A5
TCF4
CAPG
VPREB1
Hematopoiesis · GMP
RNASE2
RAB32
LYZ
CST7
CEBPD
SRGN
LYST
BZRAP1
CFD
Hematopoiesis · MLP
DNTT
CYTH4
DDIT4
SEPT1
MEF2C
LTB
RCSD1
SPINK2
CD52
Paul15 · Ery
Cldn13
Isg20
Fam132a
Ccne1
Klf1
Sphk1
Mt2
Stard10
Hemgn
target expression change, early → late (log)
activating edge
repressive edge